Vel Tech
Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
University
(Estd. u/s 3 of UGC Act, 1956)
Avadi, Chennai

NAAC A++

CATEGORY-1 DEEMED TO BE UNIVERSITY by the UGC

SCHOOL OF COMPUTING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ASSIGNMENT

Name : M. Manoj kumar

VTU Number : 29576

Course Code/ Course Name : **10212CS211/Artificial Intelligence Techniques**

Year : Summer 2026-27

Course Handling Faculty : DR. PRAGADEESWARAN.

Slot : S6 L4

Course Coordinator : Dr. S.Pragadeeswaran & TTS3903

Assignment Group Number : 9

Date : 29/10/2026

Assignment Mark Allocation

Criteria	Total Mark	Mark Achieved
Concept Understanding (A)	1	1
Accuracy & Correctness (B)	1	1
Presentation Clarity & Communication (C)	1	1
Relevance & Completion (D)	1	1
Timeliness (E)	1	1
Total (A+B+C+D+E)	5	5

General Comments:

Refer Reference Book and Ref material

Proof 28/10/26

Faculty Signature

Faculty Name & TTS:

Dr. S Pragadeeswaran
TTS: 3903

	Course Outcomes	Level of Learning domain(Based on revised Blooms taxonomy)
	Identify the problem-solving techniques by using State Space search and heuristic search approach	K3
2	Use evolutionary and advanced search techniques to provide optimized solution.	K3
3	Solve unified planning approach and Constraint-satisfaction problems.	K3
4	Apply Logical knowledge representation for rules and fact-based approach.	K3
5	Build the logic of Uncertainty and Reasoning using resolution methods.	K3

stion Group:

GROUP -1

List of Questions	CO	K Level
Formulate the given tic tac toe problem using heuristic function.	CO1	K2
Apply A* Algorithm to find the shortest path from source node A	CO2	K3
Consider the problem of Constructing (not solving) Crossword puzzle: Given a list of words	CO3	K2
Translate each of the following sentences into a conceptual graph.	CO4	K2

mulate the given tic toe problem using heur
function.

O		X
X		
X	O	O

Heuristic function: A heuristic function estimates
a game state is for a player.
Let h(n) be the heuristic value of a board st

$$h(n) = 3X_2 + X_1 - 3O_2 - O_1$$

where:

X_2 = number of winning lines with two X
one blank.

X_1 = number of winning lines one X a
blanks.

O_2 = number of winning lines with
and one blank.

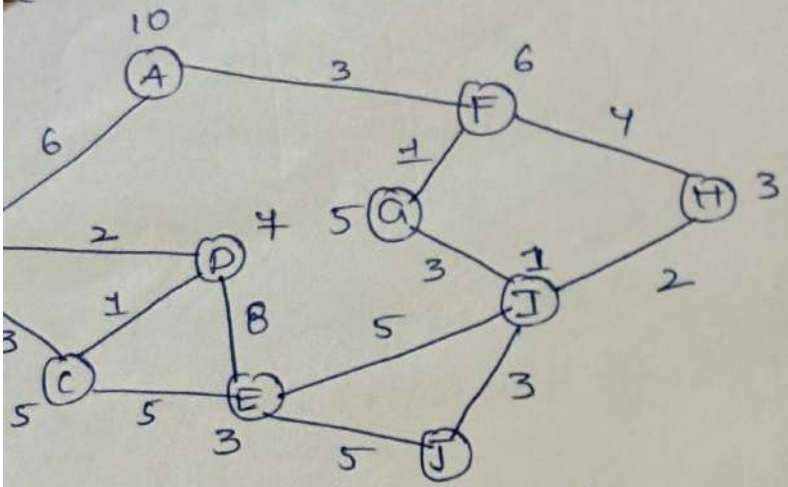
O_1 = number of winning with one O a
blanks.

mulation:

Initial state: Empty 3x3 board.

Operators: place X or O is an empty cell

A* Algorithm to find the shortest path from
ce node A to goal node J.



the evaluation function:

$$f(n) = g(n) + h(n)$$

e:
n) = Actual cost from source A to node n.
n) = Estimated cost from node n to goal J.
n) = Estimated total path cost.

hm:
t with source node A.
lculate $f(n) = g(n) + h(n)$ for the available n
ect the node with the lowest $f(n)$.
pand that node and calculate the costs
uccessors.
epeat until goal J is reached.
ace back the patent nodes to obtain th
ath.

the connection. ed

Consider the problem of constructing (not solvi
Crossword puzzles: 5 fitting words into a recta
grid. The grid, which is given as part of the
specifies which squares are blank and which
shaded. Assume list of words.

a) As a general search problem.

Initial state: the crossword grid contains blank o
squares.

State: A partially or completely filled crossword

operators: Insert a word from the dictionary in
suitable blank slot.

Goal test: All blank slots are filled with valid w
all intersecting letters match.

Cost: Each word placement has a cost of 1.

Heuristic function:

h(n) = Number of unfilled word slots.

estimates the remaining work needed to comp
crossword.

Suitable algorithm: Back-tracking search or A

b) As a Constraint Satisfaction problem (CSP)

A CSP consists of variables, domains, and c

. Variables: Each horizontal or vertical word
length



2. Word length must match the slot length
2. Intersecting words must have matching letters.
3. Words must belong to the given dictionary.

4) Translate the Sentences into Conceptual graph
1. Jane gave Tom an ice cream.
2. Basketball players are tall.
3. paul cut down the tree with an axe.
4. plate all the Ingredients in a bowl and mix th

1) [person: none]
↓ AGENT
[Give]
[person: Tom] [Ice cream]
↑ RECIPIENT ↑ THEME

2) Basketball players are tall.
[Basketball player]
↓ IS-A / Has - PROPERTY
[Tall]

[person: paul]
↓
[Cutdown]
[Tree] [Axe]
↑ ↑ INSTRUMENT
object

[person: you/ implied]

...ply resolution refutation and prove KB ⊨ Q

Knowledge Base (KB):

1. persons who consume neither nor fish are vegeta
2. Vegans consume no animal food and no milk
3. Meat is animal food.
4. Fish is animal food.

Goal: prove that vegans are vegetarian.

$C_1 : \neg M(x) \wedge \neg F(x) \rightarrow Veg(x)$

$C_2 : V(x) \rightarrow \neg M(x)$

$C_3 : V(x) \rightarrow \neg F(x)$

$C_1 : M(x) \vee F(x) \vee Veg(x)$

$C_2 : \neg V(x) \vee \neg M(x)$

$C_3 : \neg V(x) \vee \neg F(x)$

The goal $Q : V(x) \rightarrow Veg(x)$

Negating $\neg Q : V(a) \wedge \rightarrow \neg Veg(a)$

$C_4 : V(a)$

$C_5 : \neg Veg(a)$

$$\frac{\neg V(a) \vee \neg M(a), \; V(a)}{\neg M(a)}$$

C_3 with C_4 :

$$\frac{\neg V(a) \vee \neg F(a), \; V(a)}{\neg F(a)}$$

$$\frac{Veg(a), \; \neg Veg(a)}{\square}$$

$KB \not\models Q$